Chapter 4 Statically indeterminate problems

4.1 Introduction

In statics we encountered problems such as Fig. 1a. In which, the beam is supported by a pin at point A and a roller at point B.

The pin provides the support to the beam in horizontal and vertical directions. Therefore, there are external reactions by the pin at point A, R_1 and R_2.

The roller provides the support to the beam in vertical direction only. Therefore, there is external reaction by the roller at point B, R_3, in vertical direction.

To find the external reactions under this beam, we drew a free body diagram and used the equilibrium equations:

$$\sum M_A = 0,\ 2R_3 - 1.5\times 60 = 0,\ R_3 = 45\ \text{N} \qquad (4.1)$$

$$\sum F \rightarrow = 0,\ 20 - R_2 = 0,\ R_2 = 20\ \text{N} \qquad (4.2)$$

$$\sum F \uparrow = 0,\ R_1 + R_3 - 60 = 0,\ R_1 = 15\ \text{N} \qquad (4.3)$$

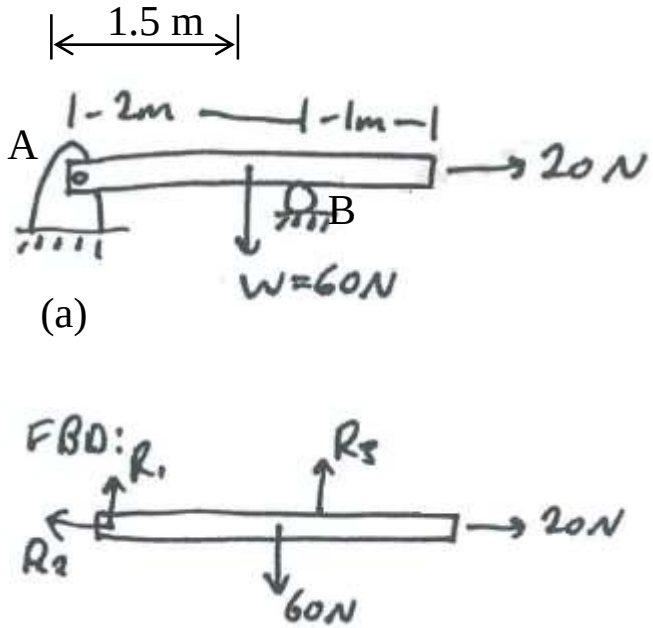


Fig. 1
(a) A statically determinate beam;
(b) Free-body-diagram

We saw the number of unknowns and the number of available equilibrium equations are equal.

The reactions could be determined solely based on the equilibrium equations.

The problems of such nature were called ***Statically Determinate*.**

We now look at the problem of Fig. 2(a). There is a roller support at the right end of the beam:

The roller at the right end provides the support to the beam in vertical direction. Therefore, there is external reaction by the roller, R_4, in vertical direction.

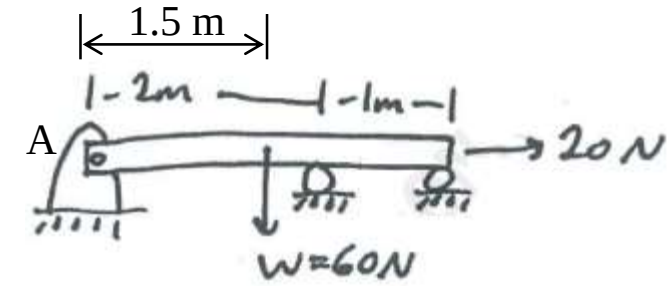
We again draw a free body diagram and use the equilibrium equations:

$$\sum M_A = 0,\ 2R_3 + 3R_4 - 1.5 \times 60 = 0 \qquad (4.4)$$

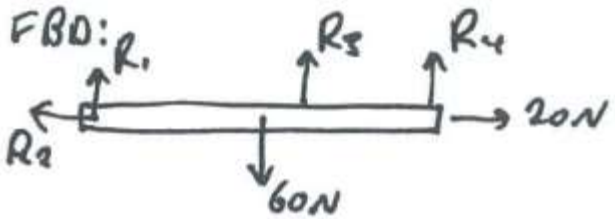
$$\sum F \rightarrow = 0,\ 20 - R_2 = 0,\ R_2 = 20\ \text{N} \qquad (4.5)$$

$$\sum F \uparrow = 0,\ R_1 + R_3 + R_4 - 60 = 0 \qquad (4.6)$$

Equations (4.4) and (4.6) are two equations in three unknowns and therefore, they can not be solved. We say that we need more information than statics can provide, and so this problem is called ***Statically Indeterminate*****.**



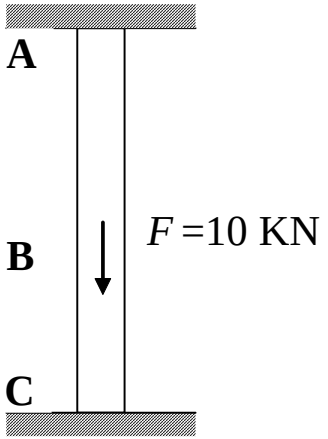
(a)

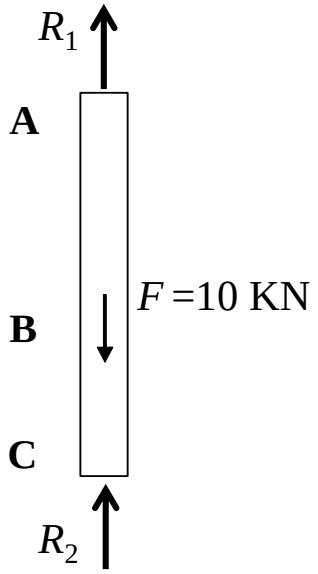


(b)

Fig. 2
(a) A statically indeterminate beam;
(b) Free-body-diagram

A similar problem is a column supported at both ends. We are required to find the unknown reactions at A and C.





From the ***equilibrium*** in the vertical direction,

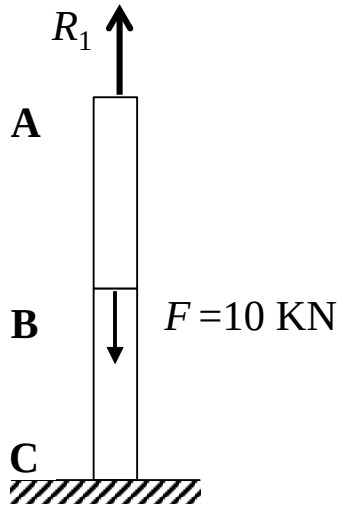
$$\sum F \uparrow = 0,\ R_1 + R_2 - F = 0 \quad (4.7)$$

This equation has two unknowns (R_1 and R_2). It therefore cannot be solved, and so this problem is also **statically indeterminate.**

To solve this problem, we start by considering an equivalent problem. This involves the same column, but we replace the unknown reaction at the top by an applied force of unknown magnitude plus a ***compatibility condition*** on the displacement (this is, the geometry of deformation) at the top.

$$\delta_{top} = 0$$

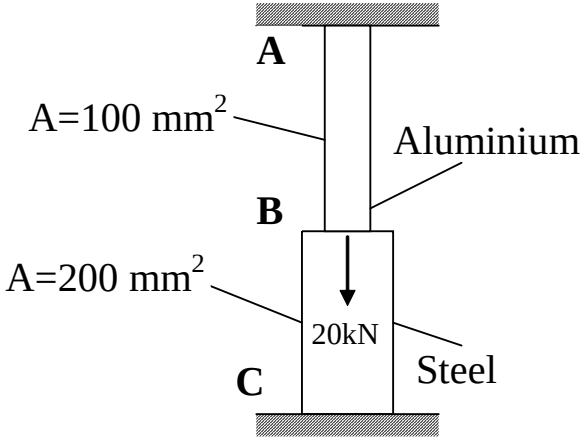
The compatibility condition makes the two problem the same as each other.

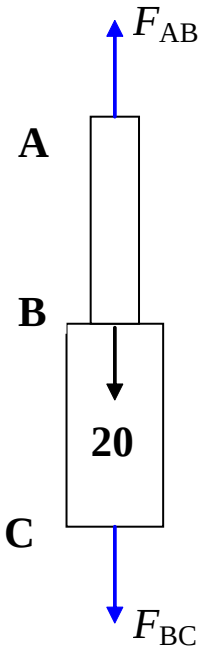


If we know the value of R_1, we could use the ***material behavior*** (stress-strain relation in Section 3.3 of Chapter 3) to determine the displacement of the top of the column.

We could then set this displacement to zero, as required by the compatibility condition, and use this to find the required value of R_1.

Example 1: Determine the reactions at the floor and ceiling (right figure). Lengths: AB = BC = 100 mm, E_{al} = 70 GPa, E_{st} = 200 GPa. Also obtain the stresses in AB and BC.





Note: F_{AB} and F_{BC} are internal forces. They are assumed to be positive (tensile). Therefore, in this figure, F_{AB} is pointing upward and F_{BC} is pointing downward.

Solution:

Step 1: Use the free body diagram to obtain the ***equilibrium condition(s)***

$$\sum F_V = 0 \quad \underset{\text{+vertical}}{\uparrow} \qquad \boxed{F_{AB} - F_{BC} - 20{,}000 = 0} \qquad (4.8)$$

Step 2: Write the ***compatibility condition*** (in this problem, the total elongation of bar AB and Bar BC are zero)

$$\delta_{AB} + \delta_{BC} = 0 \qquad (4.9)$$

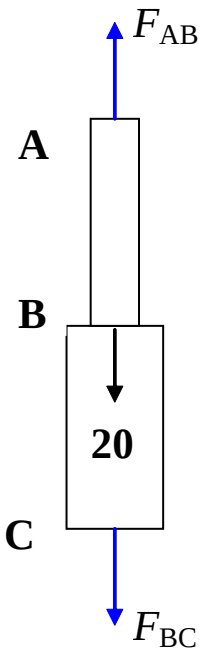
Step 3: Use the ***material behavior*** (relations between the internal forces and the elongations) of the rod (Section 3.3 of Chapter 3).

$$\boxed{\delta_{AB} = \frac{F_{AB} \cdot L_{AB}}{E_{AB} \cdot A_{AB}},\ \delta_{BC} = \frac{F_{BC} \cdot L_{BC}}{E_{BC} \cdot A_{BC}}} \qquad (4.10)$$

Step 4: Find the internal forces

Substituting Eqs. (4.8) and (4.10) into Eq.(4.9) gives

$$\frac{F_{\text{AB}} \cdot L_{AB}}{E_{AB} \cdot A_{AB}} + \frac{(F_{\text{AB}} - 20{,}000) \cdot L_{BC}}{E_{BC} \cdot A_{BC}} = 0 \qquad (4.11)$$



$$\Rightarrow \quad \frac{F_{\text{AB}} \times 0.1}{(70 \times 10^{9}) \times (100 \times 10^{-6})} + \frac{(F_{\text{AB}} - 20{,}000) \times 0.1}{(200 \times 10^{9}) \times (200 \times 10^{-6})} = 0$$

$$\Rightarrow \quad F_{\text{AB}} = 2{,}978\ \text{N} = 2.978\ \text{kN}$$

$$\Rightarrow \quad F_{\text{BC}} = -17{,}022\ \text{N} = -17.022\ \text{kN}$$

We see that:

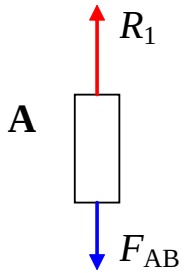
- F_{AB} is positive. This means that bar AB is in tension.
- F_{BC} is negative. Therefore, bar BC is in compression.

Note: We draw the unknown internal force F_{BC} in the positive direction (tensile). The negative value we obtain indicate that it acts compressive.

Step 5: Find the reactions – using free body diagrams

For section AB:

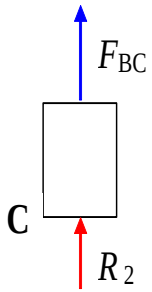
$$\sum F_V = 0 \quad \uparrow \text{+vertical}$$



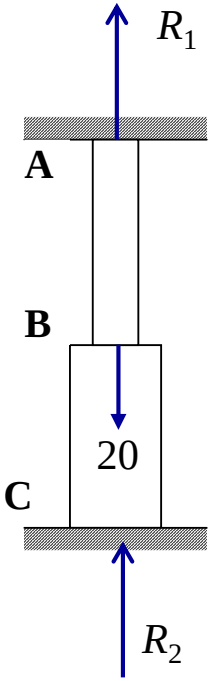
$$R_1 = F_{AB}$$

For section BC:

$$\sum F_V = 0 \quad \uparrow \text{+vertical}$$



$$R_2 = -F_{BC}$$



$$\Rightarrow \quad R_1 = 2.978\,\text{kN}$$

$$R_2 = 17.022\,\text{kN}$$

Thus, the ceiling applies a force 2.978 kN (upward) to the bar, and

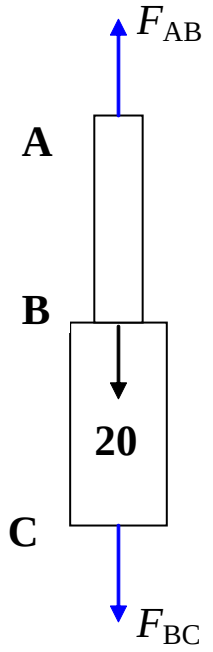
the floor applies a force 17.022 kN (upward) to the bar.

Step 6: Stress

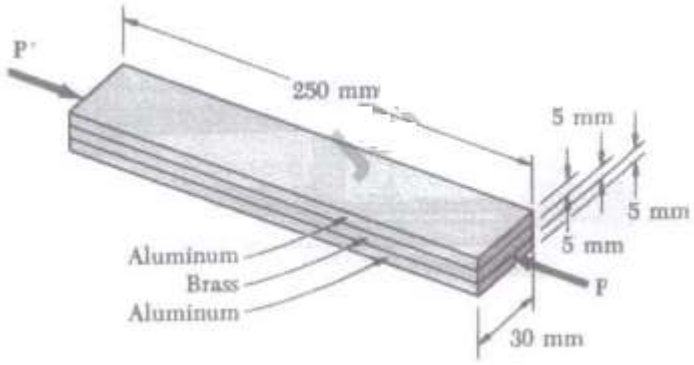
Average normal stress in sections AB and BC:

$$\sigma_{AB} = \frac{F_{AB}}{A_{AB}} = \frac{2978}{100 \times 10^{-6}} = 29.78\,\text{MPa}$$

$$\sigma_{BC} = \frac{F_{BC}}{A_{BC}} = \frac{-17{,}022}{200 \times 10^{-6}} = -85.11\,\text{MPa}$$

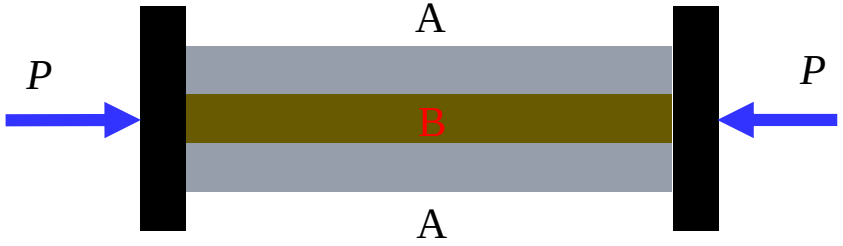


4.2 Parallel statically indeterminate problems

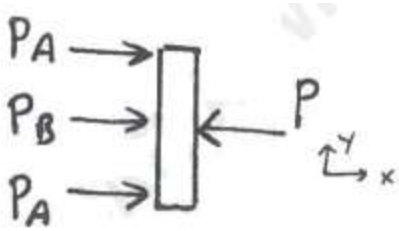


In the previous problem, the different sections were located end-to-end.

A compound bar consisting of components located parallel to each other can also be statically indeterminate.

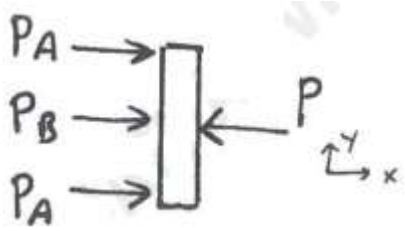


We know that the total force being carried by all components is equal to P.



However, we have no way of telling how much of the total force P is being carried by the Aluminium (P_A) strips and how much by the Brass (P_B) strip.

To solve this type of problem, we can start by imagining that an endplate is being used to distribute the applied force, and draw a free body diagram of this endplate:



From equilibrium,

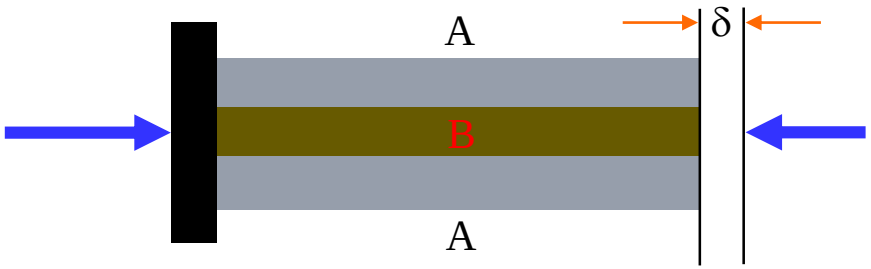
$$\sum F_x = 0 \quad \underset{+x}{\rightarrow} \quad \Rightarrow \quad P_A + P_B + P_A - P = 0$$

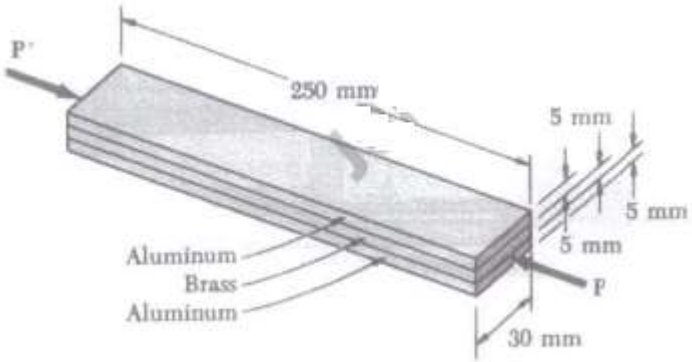
Equilibrium only gives us one usable equation relating these two unknowns:

$$2P_A + P_B - P = 0$$

The problem is therefore statically indeterminate.

In this situation, to obtain a further equation, we use the compatibility condition that the brass and the aluminum strips must both deform by the same amount.

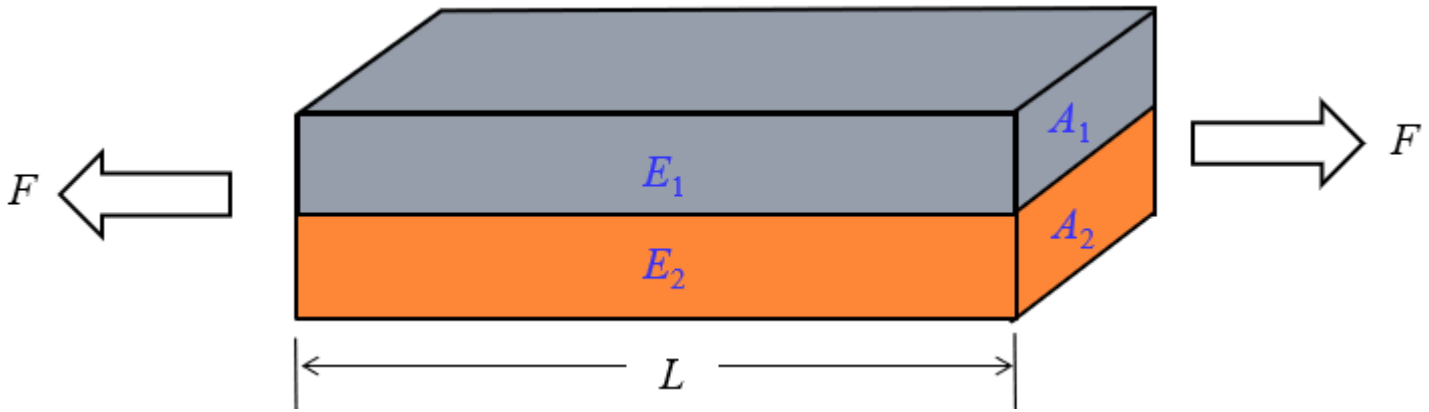


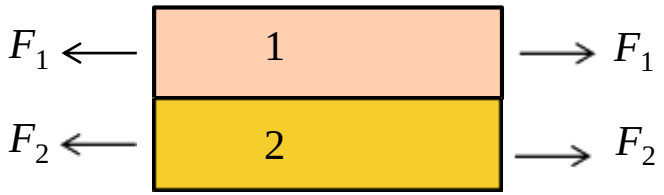


Example 2: Composite materials: A system of multiply materials under tension. The two layers are jointed securely at their ends. Properties of the layers are:
Young's modulus E_1 and cross-section area A_1 for material 1, and Young's modulus E_2 and cross-section area A_2 for material 2. The subscripts 1 and 2 denote, respectively, the materials 1 and 2. Determine

(a) the elongation δ_0 of the layers;

(b) the axial stress in each layer.





Solution: This is a statically indeterminate problem.

(1) Equilibrium equation: $F_1 + F_2 = F$

(2) Compatibility equation: elongations of the two layers are equal, $\delta_1 = \delta_2$. Denote them as δ_0

(3) Material behavior: $\delta_1 = \frac{F_1 L}{E_1 A_1}$, $\delta_2 = \frac{F_2 L}{E_2 A_2}$

(4) From **(1) - (3)**,

(a) $\delta_1 = \delta_2 = \delta_0 = \frac{FL}{E_1 A_1 + E_2 A_2}$

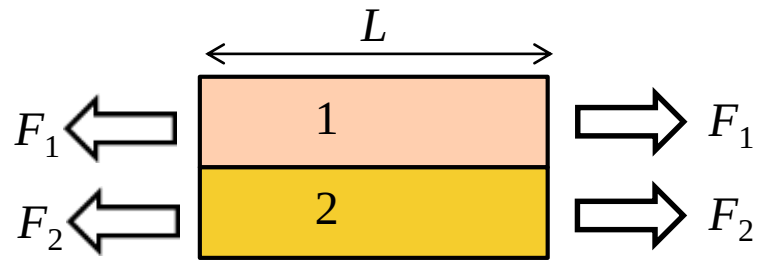
(b) To find the stresses, we need to find the internal forces first.

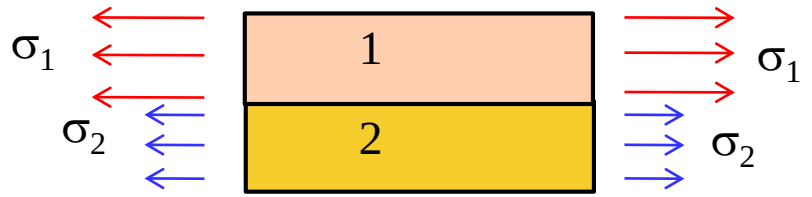
Putting (a) into (3) gives:

$$F_1 = \frac{E_1 A_1}{E_1 A_1 + E_2 A_2} F \quad , \quad F_2 = \frac{E_2 A_2}{E_1 A_1 + E_2 A_2} F$$

The stress in each layer:

$$\sigma_1 = \frac{F_1}{A_1} = \frac{E_1 F}{E_1 A_1 + E_2 A_2} \quad , \quad \sigma_2 = \frac{F_2}{A_2} = \frac{E_2 F}{E_1 A_1 + E_2 A_2}$$





Subscripts 1 and 2 denote material 1 and material 2

4.3 Note:

When solving the statically indeterminate problems, three fundamental concepts must be use:

(1) The ***equilibrium conditions*** (force balance) must be satisfied.

(2) The ***compatibility condition*** (geometry of deformation) must be described.

(3) The ***material behavior*** (i.e., the force-deformation relation of the materials) must be characterized.